Vishnu Panganamamula . Jarvis Consulting

Hello! I am Vishnu Panganamamula and I am a recent computer engineering graduate (June 2025) from Toronto Metropolitan University (formerly Ryerson University). I have hands-on experience working with full-stack development, cloud services and AI/ML. I currently work at Jarvis as a data engineer, where I work on projects that will prepare me for industry level work in the future. I am eager to contribute to meaningful projects where I can use my problem-solving and programming skills to create impactful solutions.

Skills

Proficient: Java, Linux/Bash, RDBMS/SQL, Agile/Scrum, Git

Competent: Python, Docker, Tensorflow, PyTorch, Google Cloud/Azure

Familiar: React.js, Javascript, Kubernetes, C/C++, Firebase

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_VishnuPanganamamula

Cluster Monitor [GitHub]: Used Linux commands, Bash scripting, PostgreSQL database and Docker to create a Linux Cluster Monitoring Tool. The monitoring system records specific hardware information of a host VM and what resources of the system are being used. This usage data is recorded and monitored from a host VM running Rocky Linux in real time. The data for the host usage is collected and stored in the PostgreSQL database regularly, allowing users to check their VM usage activity.

Core Java Apps [GitHub]:

- Twitter App: Curabitur laoreet tristique leo, eget suscipit nisi. Sed in sodales ex. Maecenas vitae tincidunt dui, et eleifend quam.
- JDBC App: Curabitur laoreet tristique leo, eget suscipit nisi. Sed in sodales ex. Maecenas vitae tincidunt dui, et eleifend quam.
- Grep App: Curabitur laoreet tristique leo, eget suscipit nisi. Sed in sodales ex. Maecenas vitae tincidunt dui, et eleifend quam.

Springboot App [GitHub]: Not Started

Python Data Analytics [GitHub]: Not Started

Hadoop [GitHub]: Not Started

Spark [GitHub]: Not Started

Cloud/DevOps [GitHub]: Not Started

Highlighted Projects

Societal Unrest Prediction Model [GitHub]: Developed logistic regression model and random forest classifier to predict monthly unrest events across 50 global regions based on 10 years of economic, environmental, and sociopolitical indicators Engineered temporal and interaction-based features and applied time-series cross-validation to improve recall from <5% to ~95%while maintaining high precision and overall model performance (F1, AUC-ROC > 90%) Proposed actionable policy analyst alert rules based on feature importances and output probability thresholds

Medical Imaging Diagnosis Tool [GitHub]: Led model development for lung disease classification using CNNs (ResNet-50, EfficientNet-B4, DenseNet-121) and the NIH Chest X-rays Dataset (~112k images) Designed and tested a custom CNN (based on LeNet) with batch normalization and BCEWithLogitsLoss; results informed shift to more effective pre-trained models Implemented pre-trained U-Net for disease segmentation and for visual purposes, improving interpretability and localization of model predictions Built and optimized data pipelines using tf.data for efficient data loading and training performance Contributed to model selection, hyperparameter tuning and performance evaluation of models Trained and optimized models to reach up to 90%+ prediction accuracy for lung diseases

Professional Experiences

Data Engineer, Jarvis (July 2025-present): Developing industry related projects within a dynamic Agile/Scrum environment.

AI Software Developer Intern, Vosyn Inc. (Sept. 2023-Dec. 2023): Spearheaded research on LLMs, TTS and STT systems, contributing key insights that shaped software architecture decisions for Vosyn's AI platforms Fine-tuned the Bark TTS model to enhance multilingual accuracy and speech-to-text reliability, improving model performance across diverse languages Customized Whisper AI for Japanese audio-to-text translation, increasing transcription accuracy through iterative testing on varied audio outputs Designed and implemented a robust data pipeline model and database system and backend for VosynVerse project with PostgreSQL and Django, supporting seamless data integration and scalability Conducted risk assessments and safety documentation for Vosyn LLM features, proactively identifying development bottlenecks and offering mitigation strategies

Education

Toronto Metropolitan University (Sept. 2020-June 2025), Bachelor of Engineering, Electrical and Computer Engineering - Dean's List (Fall 2024): Earned Dean's List award for Fall 2024 for outstanding academic excellence

Miscellaneous

- Udacity Machine Learning (2019)
- Winner
- Combat Sports (Boxing)
- Casual Gamer
- Drummer